Mingyun Kang

Researcher in Construction Management & Automation

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Education

Seoul National University of Science and Technology — *Seoul, Republic of Korea*
B.S. in Architectural Engineering, School of Architecture — *Mar 2017 – Feb 2024*
- Double Major: Business Administration
- GPA : 3.8 / 4.5

Seoul National University of Science and Technology — *Seoul, Republic of Korea*
M.S. in Architectural Engineering (Integrated Bachelor-Master Program) — *Mar 2024 – Aug 2025*
- GPA : 4.22 /4.5
- Thesis : Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System (Advisor : Prof. Taehoon Kim)

Research Experience

Research Assistant — *Seoul, Republic of Korea*
CONICT Lab, Seoul National University of Science and Technology — *Dec 2021 – Aug 2025*
Conducted research on computer vision-based automated construction supervision, 3D reconstruction, and human-robot collaboration at construction sites.
- Published 3 SCIE journal papers and 2 SCOPUS conference papers as first/co-author
- Filed 5 domestic patent applications related to construction automation and vision-based inspection
- Participated in 3 national R&D projects funded by NRF and the Ministry of Land, Infrastructure and Transport (MOLIT)

Post-Master's Researcher — *Goyang, Republic of Korea*
Korea Institute of Civil Engineering and Building Technology (KICT) — *Nov 2025 – present*
Post-Construction Evaluation Center — Systematically evaluating outcomes of publicly funded SOC
- Compliance & Implementation Support: Developing research frameworks to encourage clients to fulfill post-construction evaluation requirements and to facilitate smooth reporting workflows.
- Feedback Loop & Knowledge Management: Designing mechanisms to extract structured insights from collected evaluation data, enabling lessons learned from completed projects to feed back into future initiatives and build a virtuous cycle of construction knowledge.

R&D Projects

Digital Mapping-based Platform Research Laboratory for Healthy Construction Lifecycle — *June 2022 – Feb 2025*
National R&D project exploring digital mapping-based platforms for the full construction lifecycle. Focused on computer vision applications for automated construction supervision.

Development of Digital-based Architectural Supervision and Construction Automation Robot Technology — *Apr 2022 – Dec 2026*
Large-scale national R&D project (KRW 24,490M) funded by MOLIT aimed at developing digital architectural supervision systems and construction automation robots.

Development of Human-Robot Collaboration Technology and Integrated Operation System for High-Altitude Work at Construction Sites — *Apr 2025 – Dec 2027*
Ongoing national R&D project (KRW 17,000M) funded by MOLIT developing XR-based human-robot collaboration platforms for multipurpose high-altitude construction work.

Publications (SCIE)

Computer Vision-Based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System Dec 2024
Applied Sciences, 15(1), 125.
Mingyun Kang, Sebeen Yoon, Taehoon Kim
10.3390/app15010125 (Applied Sciences)

Approach to Enhancing Panoramic Segmentation in Indoor Construction Sites Based on a Perspective Image Segmentation Foundation Model Apr 2025
Applied Sciences, 15(9), 4875.
Juho Han, Sebeen Yoon, **Mingyun Kang**, Taehoon Kim
10.3390/app15094875 (Applied Sciences)

Automated Vision-based Location Monitoring for Jack Support Jan 2025
KSCE Journal of Civil Engineering, 100374.
Sebeen Yoon, **Mingyun Kang**, Minhyung Kim, Taehoon Kim
10.1016/j.kscej.2025.100374 (KSCE Journal of Civil Engineering)

Publications (SCOPUS)

Instance Segmentation of Exterior Insulation Finishing System using Synthetic Datasets June 2024
ISARC Vol. 41, pp. 1176-1181. IAARC Publications.
Mingyun Kang, Sebeen Yoon, Juho Han, Sanghyeon Na, Taehoon Kim
10.22260/ISARC2024/0152 (ISARC 2024 — International Symposium on Automation and Robotics in Construction)

Comparative Study of Structure from Motion on Construction Site July 2025
ISARC Vol. 42, pp. 1395-1400. IAARC Publications.
Mingyun Kang, Sangmin Lee, Sebeen Yoon, Taehoon Kim
10.22260/ISARC2025/0180 (ISARC 2025 — International Symposium on Automation and Robotics in Construction)

Publications (International Conference)

Computer Vision-Based Tile Counting Model for Automated Progress Monitoring June 2023
CCC 2023, Keszthely, Hungary.
Mingyun Kang, Wooseok Lee, Junsang Yoo, Sebeen Yoon, Taehoon Kim
(Creative Construction Conference (CCC 2023))

A Preliminary Study on Automatic Jack Support Installation Interval Measurement Model June 2023
CCC 2023, Keszthely, Hungary.
Sebeen Yoon, **Mingyun Kang**, Taehoon Kim
(Creative Construction Conference (CCC 2023))

Approach for Improving Panoramic Image Segmentation Quality on Construction Sites using SAM June 2024
CCC 2024, Prague, Czech Republic.
Juho Han, Sanghyeon Na, **Mingyun Kang**, Sebeen Yoon, Taehoon Kim
(Creative Construction Conference (CCC 2024))

Publications (Domestic)

Computer Vision-based Adhesion Quality Inspection Automation Model for Exterior Insulation and Finishing System Mar 2023
JKIBC, 23(2), 165-173. (KCI)
Sebeen Yoon, **Mingyun Kang**, Hyeonseung Jang, Taehoon Kim
(Journal of the Korean Institute of Building Construction (JKIBC))

Applicability Enhancement of Deep Learning-based Computer Vision Models for 360° Panoramic Images Nov 2023
KICEM 2023, Goseong, 205-206.
Mingyun Kang, Junsang Yoo, Wooseok Lee, Taehoon Kim
(Proceedings of KICEM Annual Conference)

Performance Comparison Study on Image-based 3D Space Generation Oct 2024
AIK Fall 2024, Gyeongju, 974-975.
Mingyun Kang, Sebeen Yoon, Taehoon Kim
(Architectural Institute of Korea Fall Conference 2024)

A Basic Study on 3D Reconstruction Performance Improvement Using 360° Camera Nov 2024
KIBC Autumn 2024, 203-204.
Mingyun Kang, Juho Han, Sebeen Yoon, Taehoon Kim
(Proceedings of KIBC Autumn Conference)

Patents

Automatic Supervision Method and Device for Adhesive Application Quality Apr 2023
Domestic Patent Application No. 10-2023-0053756. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**

Patents (Pending)

Automated Construction Site Supervision System Sept 2023
Domestic Patent Application No. 10-2023-0125464. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**

Tile Quantity Counting System Oct 2023
Domestic Patent Application No. 10-2023-0135013. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**, Wooseok Lee, Junsang Yoo

Method for Measuring Rebar Spacing Based on Single Image Information Sept 2024
Domestic Patent Application No. 10-2024-0127210. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, **Mingyun Kang**, Cheolhun Park, Jihyeon Lee, Jieun Lee, Yunjae Jeon

Method for Detecting Illegal Rooftop Buildings Based on Video Image Analysis Aug 2025
Domestic Patent Application No. 10-2025-0117608. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.
Taehoon Kim, **Mingyun Kang**, Seonwoo Lee, Yeon Choi, Yujin Oh, Gangmin Bae

Awards

Best Paper Award May 2023
Awarded at the Spring Conference 2023 for the paper: A Basic Study on Image-based Numerical Measurement Model for Construction Sites
Korean Institute of Building Construction (KIBC)

Best Paper Award May 2024
Awarded at the Spring Conference 2024 for the paper: A Basic Study on Rebar Spacing Measurement Method Based on Single Image Information
Korean Institute of Building Construction (KIBC)

Excellence Paper Award May 2025
Awarded at the Spring Conference 2025 for the paper: A Basic Study on Image Analysis-based Crackdown on Illegal Buildings on Rooftops
Korean Institute of Building Construction (KIBC)

Outstanding Thesis Award Aug 2025
Awarded for the Master's thesis: Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System
Seoul National University of Science and Technology

Languages & Skills

- Korean (Native) · English (Advanced) · Python (Advanced)

Certificates

Architectural Engineer License *June 2025*

TOEFL iBT — Score 4.5/6.0 (90/120) *July 2026*
RC 5.0 / LC 5.0 / WRT 3.5 / SPK 4.5

References

Professor Taehoon Kim